

BIOPYTHON PIPELINE

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Python 3.13 (64-bit)
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> # Import Libraries
>>> from Bio import Entrez
>>> from Bio import SeqIO
>>> from Bio.SeqUtils import gc_fraction
>>> # Set up Email
>>> Entrez.email = "shreyadungdung18@gmail.com"
>>> handle = Entrez.efetch(
...     db="nucleotide",
...     id="NM_000546",
...     rettype="fasta",
...     retmode="text"
... )
>>> record = SeqIO.read(handle, "fasta")
>>> handle.close()
>>> # Display information
>>> print(record.id)
NM_000546.6
>>> print(record.description)
NM_000546.6 Homo sapiens tumor protein p53 (TP53), transcript variant 1, mRNA
>>> print(len(record.seq))
2512
>>> # Calculate GC Content
>>> gc = gc_fraction(record.seq)*100
>>> print(f"GC Content: {gc:.2f}%")
GC Content: 53.38%
>>> # Translate DNA
>>> protein = record.seq.translate()
C:\Users\HP\AppData\Local\Programs\Python\Python313\Lib\site-packages\Bio\Seq.py:2874: BiopythonWarning: Partial codon, len(sequence) not a multiple of three. Explicitly trim the sequence or add trailing N before translation. This may become an error in future.
  warnings.warn(
>>> print(protein[:60])
LKSLPPSREQVAAGLRGHFAFGLGACFPRR*HASLDWQPDCLPGHCHGGAIVRS*RRAP

>>> # Saving the results
>>> with open("tp53_report.txt", "w") as file:
...     file.write(f"Gene: {record.description}\n")
...     file.write(f"Length: {len(record.seq)}\n")
...     file.write(f"GC Content: {gc:.2f}%\n")
...     file.write(f"\nProtein:\n")
...     file.write(str(protein))
...
84
13
19
10
837
```